ADITI MISHRA

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SUMMARY

Senior Researcher working at Fujitsu Research Human-Computer Interaction focusing on multimodal AI systems, video understanding, and human-centered interaction systems.

EDUCATION

Arizona State University *June 2019 - 2024*
Doctor of Philosophy (PhD) *Current GPA: 3.82/4*
Major: Computer Science

International Institute of Information Technology - Bhubaneswar *August 2014 - May 2018*
Bachelor of Technology (BTech). CGPA: 8.53/10
Major: Computer Science and Engineering

PUBLICATIONS

- Jo, H.Y., Sakashita, M., **Mishra, A**, Niinuma, K., Gupta, A. Map2Vid: Street View Driven AI Video Generation, *ACM User Interface Software and Technology, 2026*
- **Mishra, A.**, Niinuma, K., Gupta, A. FuzzySeek: Multimodal Refinement of Imprecise Queries for Video Moment Retrieval, *ACM Intelligent User Interfaces, 2026*
- Hu, X., Sakashita, M., **Mishra, A**, Niinuma, K., Gupta, A. SGMotion: Scene-Grounded Motion Authoring for Human–Scene Interactions via Video Generation, *Submitted to UIST, 2026*
- **Mishra, A.**, Brudy, F., Zhou, Q., Fitzmaurice, G., Anderson, F. WhatIF: Branched Narrative Fiction Visualization for Authoring Emergent Narratives using Large Language Models, *C&C, 2025*, **Honorable Mention Award**
- **Mishra, A.**, Biswas, A., Bryan, C. VeCNA: Visual Exploration, Comparison and Analysis of Reconstructed Spatiotemporal Scientific Simulation Data, *EuroVis Short Papers, 2025*
- **Mishra, A.**, Danzy, B. , Soni, U., Arunkumar, A., Huang, J., Kwon,B.C., Bryan, C. PromptAid: Prompt Exploration, Perturbation, Testing and Iteration using Visual Analytics for Large Language Models, *TVCG, 2025*
- Biswas, A., **Mishra, A.**., Majumder, M., Hazarika, S., Most, A., Castorena, J., Bryan, C., McCormick, P., Ahrens, J., Lawrence, E., Hagberg, A. Filling the Void: Data-Driven Machine Learning-Based Reconstruction of Sampled Spatiotemporal Scientific Simulation Data ,*DRBSD 2024* **(link)**
- Huang, J., Chen, C. **Mishra, A**, Kwon,B.C., Liu, Z., Bryan, C. On CILP's Ability of Analyzing Fake Images at Large Scale: Why are they fake? *GenAICHI, 2024*
- **Mishra, A.**, Rahman, S., Kim, H., Mitra, K., Hruschka, E. Characterizing Large Language Models as Rationalizers of Knowledge-intensive Tasks ,*ACL 2024* **(link)**
- Huang, J., **Mishra, A.**, Kwon,B.C., Bryan, C. ConceptExplainer: Understanding the Mental Model of Deep Learning Algorithms via Interactive Concept-based Explanations. *IEEE Vis 2022* **(link)**

- **Mishra, A.**, Soni, U., Huang, J., Bryan, C. Why? Why not? When? Visual Explanations of Agent Behavior in Reinforcement Learning. *IEEE 15th Pacific Visualization Symposium (PacificVis), 2022* **(link)**
- **Mishra, A.**, Ginjpalli, S., Bryan, C. News Kaleidoscope: Visual Investigation of Coverage Diversity in News Event Reporting. *IEEE 15th Pacific Visualization Symposium (PacificVis), 2022* **(link)**
- Zhao, J., Xu, S., Chandrasegaran, S., Bryan, C., Du, F., **Mishra, A.**, Qian, X., Li. Y., Ma, K.-L. (2021). ChartStory: Automated Partitioning, Layout, and Captioning of Charts into Comic-Style Narratives. IEEE Transactions on Visualization and Computer Graphics (2021). **(link)**
- Bryan, C., **Mishra, A.**, Shidara, H., & Ma, K.-L. (2020). Analyzing Gaze Behavior for Text-embellished Narrative Visualizations under Different Task Scenarios. Visual Informatics, 4(3), 41–50, 2021. **(link)**
- Huang, J., **Mishra, A.**, Arunkumar, A., & Bryan, C. (2020). TotemFinder: A Visual Analytics Approach for Image-based Key Players Identification. In 2019 IEEE Conference on Visual Analytics Science and Technology (VAST). **VAST Challenge 2019 Honorable Mention**.
- **Mishra, A.**, Mallick, S. , Mukhopadhyay, R., Mukherji, K., Bryan, C. PolicySummarizer: Visual Summaries for Reinforcement Learning using Semantic Segmentation, *In Submission, Visual Informatics, 2025*

WORK EXPERIENCE

Fujitsu Research, Pittsburgh - *Senior Researcher* *Oct 2024 - Present*
Advisor: Dr Koichiro Niinuma, Dr Aakar Gupta
- Working on GenAI assisted tools to enable multimodal video understanding.

Autodesk Research, Toronto - *HCI+AI Research Intern* *May 2024 - Sept 2024*
Advisor: Dr Fraser Anderson, Dr Frederik Brudy, Dr Qian Zhou
- Working on LLM assisted authoring tools to enable authors write game play narratives.
- Building a visualization system to help game developers author and version interactive fiction.

ASU Sonoran Visualization Lab, Tempe - *Research Assistant* *Aug 2019 - Sept 2024*
Advisor: Dr Chris Bryan
- Developed an AI-driven frontend interface to recommend **prompting** changes for **RoBERTa, T0pp 5b, and LLaMA-2 13b language models**, incorporating linguistic and contextual cues.
- Reduced prompt engineering **time by 30%** and **increased task accuracy by 10%**, with a **96%** user preference.
- **Tech Stack:** Pytorch, Python, Huggingface, AWS Sagemaker, Databricks

Stanford Healthcare, Palo Alto - *Data Science Intern* *Jan 2024 - May 2024*
Advisor: Dr Nikesh Kotecha
- Working on developing tools for summarizing and evaluating longitudinal patient heath records (EHR) using **GPT-4 32k** without the presence of ground truths.
- Developing **RAG-based approaches** to medical question answering for clinicians.
- Developed and deployed a visual analytics interface to **debug and self-refine LLM chains**.
- **Tech Stack:** Pytorch, Python, Huggingface, Docker

Megagon Labs, Mountain View - *Summer Research Intern* *May 2023 - Aug 2023*
Advisor: Dr Sajjadur Rahman
- Developed a rationalization framework using **GPT-3,4** to **rationalize knowledge augmented models (KGAM)** for question answering task.
- Employed **Docker** and **AWS EC2** instances for model training and inferencing.

- Rationalizations generated **surpassed State-of-the-Art benchmark** with an approval rating of 62%.
- **Tech Stack:** Pytorch, Python, Huggingface, AWS EC2, Docker, Mechanical Turk

Los Alamos National Lab, Remote - *Summer Research Intern* *June 2020 - August 2020*
Advisor: Dr Ayan Biswas

- Built a suite of **deep learning models** to reconstruct large unstructured scientific datasets.
- Developed frontend for expert users to visually analyze differences in reconstruction qualitatively and quantitatively.
- **Tech Stack:** Pytorch, Python, D3.Js, NodeJS, Flask, React

TECHNICAL STRENGTHS

Skills: Visual Analytics, Explainable AI (XAI), Trustworthy AI, Fairness, Reinforcement Learning, Large Language Models, Deep Learning, Machine Learning, MLOps
Languages: Python, R, C, C++, MySQL, MLOps
Frameworks: Linux, PyTorch, PySpark, Langchain, Keras, Tensorflow, Docker, Django
Web Technologies: D3.js, HTML5, CSS3(Grid, Flexbox, SASS), Bootstrap, Node.js Java Script, jQuery, NodeJS, ReactJS, React Router, Flask, REST API, Svelte
Databases and Cloud Platforms: AWS, S3, SQL(Oracle, MYSQL), NoSQL (Mongo)

HONORS AND AWARDS

Special Recognition for Outstanding Review, CHI *2026*
Special Recognition for Outstanding Review, C&C *2026*
Special Recognition for Outstanding Review, UIST *2025*
Special Recognition for Outstanding Review, PacificVis *2025*
Honorable Mention, WhatIF, C&C *2025*
SCAI Doctoral Fellowship, ASU *2022*
Graduate College Travel Award, ASU *2022*
Honorable Mention, TotemFinder, IEEE VAST *2019*
Grace Hopper Scholar - Orlando, Florida *2019*

PATENTS FILED

- Techniques for implementing branched narrative fiction visualizations for authoring emergent narratives using large language models. 63/799,302
- Multimodal disambiguation of video related queries. 19/368,661
- Map2Video: Street Image Driven AI Video Generation. 63/906,389

ACADEMIC SERVICES

2027 **Program Committee**, IUI
2026 **Program Committee**, ACL
2026 **Program Committee**, IUI
2026 **Paper Reviewer**, CHI
2025 **Paper Reviewer**, UIST
2025 **Paper Reviewer**, PacificVis
2025 **Paper Reviewer**, CHI
2024 **Paper Reviewer**, IUI
2024 **Paper Reviewer**, VIS
2023 **Paper Reviewer**, VIS

MENTORSHIP EXPERIENCE

- **PhD Students -** Jason Ding (University of Maryland, College Park), (Hye-Young Jo (CU Boulder), Biswaksen Patnaik (University of Maryland, College Park), Kaustuv Mukherji (Arizona State University)
- **MS Students**
    1. Kannak Sharma, Sanidhya Chauhan, Ravindersingh Anand, Niranjan Pai, Manimozhi Shekar, Sushmita Mallick, Riyank Mukhopadhyay (Arizona State University)
- **UG Students -** Shashank Ginjpalli (Arizona State University)